

Rajalakshmi Engineering College

Name: Naren Kartic B
Email: 241901065@rajalakshmi.edu.in
Roll no:
Phone: 6374678252
Branch: REC
Department: CSE (CS) - Section 1
Batch: 2028
Degree: B.E - CSE (CS)

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 3_CY

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement:

Mason is participating in a coding challenge where he must manipulate an integer array. His task is to replace every element in the array with the next greatest element to its right. The last element of the array remains unchanged, as there is no element to its right.

Your job is to help Mason write a program that performs this transformation and outputs the modified array.

Input Format

The first line of input contains an integer n representing the number of elements in the array.

The second line of input contains n space-separated integers representing the elements of the array.

Output Format

The output prints the modified array of n integers, where each element (except the last one) is replaced by the maximum element to its right, and the last element remains unchanged.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 6

12 3 91 15 12 14

Output: 91 91 15 14 14 14

Answer

```
import java.util.Scanner;

class NextGreatestElement {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        for (int i = 0; i < n - 1; i++) {
            int max = arr[i + 1];
            for (int j = i + 1; j < n; j++) {
                if (arr[j] > max) {
                    max = arr[j];
                }
            }
            arr[i] = max;
        }

        for (int i = 0; i < n; i++) {
            System.out.print(arr[i]);
            if (i < n - 1) {
```

```
        System.out.print(" ");  
    }  
} }  
}
```

Status : Correct

Marks : 10/10

2. Problem Statement:

Imagine you have an array of integer values, and you're tasked with identifying a pair of elements within the array. This pair of elements should have a sum that is the closest to zero when compared to any other pair in the array.

Your goal is to create a program that solves this problem efficiently. The program should accept an array of integers and return the pair of elements whose sum is closest to zero.

Input Format

The first line of the input is an integer N representing the size of the array.

The second line of the input contains N space-separated integer values.

Output Format

The output is displayed in the following format:

"Pair with the sum closest to zero: {value} and {value}"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

9 10 -3 -5 -2

Output: Pair with the sum closest to zero: 9 and -5

Answer

```

import java.util.*;

class ClosestSumToZero {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Read array size
        int n = scanner.nextInt();

        // Read array elements
        int[] arr = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        // Find pair with sum closest to zero
        findClosestSumPair(arr, n);

        scanner.close();
    }

    public static void findClosestSumPair(int[] arr, int n) {
        int minSum = Integer.MAX_VALUE;
        int first = 0, second = 0;

        // Check all possible pairs
        for (int i = 0; i < n - 1; i++) {
            for (int j = i + 1; j < n; j++) {
                int sum = arr[i] + arr[j];

                // Check if this sum is closer to zero
                if (Math.abs(sum) < Math.abs(minSum)) {
                    minSum = sum;
                    first = arr[i];
                    second = arr[j];
                }
            }
        }

        // Output the result
        System.out.println("Pair with the sum closest to zero: " + first + " and " +
            second);
    }
}

```

```
}  
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Alex is a treasure hunter who collects valuable items during their quests. Each item has a specific point value, and Alex wants to maximize their score by strategically removing items one at a time.

The rule is simple: Alex removes the item with the highest point value in each step until no items are left, summing the values of the removed items to calculate the maximum score.

Help Alex to complete his task.

Input Format

The first line of input consists of an integer N, representing the size of the array.

The second line of input consists of N space-separated integers, representing the point values of the items.

Output Format

The output prints "Maximum Sum: " followed by the calculated maximum score after removing all items.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 14

7 14 21 28 35 42 49 56 63 70 77 84 91 98

Output: Maximum Sum: 735

Answer

```
import java.util.Scanner;
```

```

class MaxSumReduction {
public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);

    int size = scanner.nextInt();
    int[] arr = new int[size];

    for (int i = 0; i < size; i++) {
        arr[i] = scanner.nextInt();
    }

    int maxSum = 0;

    while (size > 0) {
        int maxIndex = 0;
        int maxElement = arr[0];

        for (int i = 1; i < size; i++) {
            if (arr[i] > maxElement) {
                maxElement = arr[i];
                maxIndex = i;
            }
        }

        maxSum += maxElement;

        for (int i = maxIndex; i < size - 1; i++) {
            arr[i] = arr[i + 1];
        }
        size--;
    }

    System.out.println("Maximum Sum: " + maxSum);
    scanner.close();
}
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Nikila is working as an intern in a software firm and is practicing with a matrix where each row represents a set of numerical values. Her task is to identify the row with the highest sum of its elements and remove that row from the matrix. After removing the row with the highest sum, Nikila needs to print the updated matrix.

Your task is to help Nikila in implementing the same. If there are two or more rows that have same the highest sum, the firstly encountered row is deleted.

Input Format

The first line of the input consists of two space-separated integers, R and C, representing the number of rows and columns in the matrix, respectively.

The following R lines each contain, C space-separated integers representing the matrix elements.

Output Format

The output prints the matrix after removing the row with the highest sum. Each row should be printed on a new line, with elements separated by a space.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 2 2

1 2

3 4

Output: 1 2

Answer

```
import java.util.Scanner;

class MatrixRowRemoval {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int rows = scanner.nextInt();
```

```

int cols = scanner.nextInt();
int[][] matrix = new int[rows][cols];

for (int i = 0; i < rows; i++) {
    for (int j = 0; j < cols; j++) {
        matrix[i][j] = scanner.nextInt();
    }
}

int maxSumRowIndex = 0;
int maxRowSum = Integer.MIN_VALUE;

for (int i = 0; i < rows; i++) {
    int rowSum = 0;
    for (int j = 0; j < cols; j++) {
        rowSum += matrix[i][j];
    }
    if (rowSum > maxRowSum) {
        maxRowSum = rowSum;
        maxSumRowIndex = i;
    }
}

int[][] newMatrix = new int[rows - 1][cols];
int newRowIndex = 0;

for (int i = 0; i < rows; i++) {
    if (i == maxSumRowIndex) {
        continue;
    }
    for (int j = 0; j < cols; j++) {
        newMatrix[newRowIndex][j] = matrix[i][j];
    }
    newRowIndex++;
}

for (int i = 0; i < rows - 1; i++) {
    for (int j = 0; j < cols; j++) {
        System.out.print(newMatrix[i][j] + " ");
    }
    System.out.println();
}

```



```
        scanner.close();  
    }  
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 3_MCQ

Attempt : 1
Total Mark : 15
Marks Obtained : 15

Section 1 : MCQ

1. What will be the output of the following code?

```
class M {  
    public static void main(String[] args) {  
        int[][] arr = {  
            {1, 2},  
            {3, 4},  
            {5, 6}  
        };  
  
        for (int i = 0; i < arr.length; i++) {  
            System.out.print(arr[i][0] + " ");  
        }  
    }  
}
```

Answer

1 3 5

Status : Correct

Marks : 1/1

2. What will be the output of the following code?

```
public class Test {  
    public static void main(String[] args) {  
        int[] x = {4, 8, 12};  
        int result = x[0] * x[2];  
        System.out.println(result);  
    }  
}
```

Answer

48

Status : Correct

Marks : 1/1

3. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[] a = {1, 2, 1, 3, 1, 4};  
        int count = 0;  
        for (int i = 0; i < a.length; i++) {  
            if (a[i] == 1) count++;  
        }  
        System.out.println(count);  
    }  
}
```

Answer

3

Status : Correct

Marks : 1/1

4. What will be the output of the given code?

```
public class Main {  
    public static void main(String[] args) {  
        int[] arr = {1, 2, 3, 4, 5};  
        int n = arr.length;  
        int temp = arr[0];  
  
        for (int i = 0; i < n - 1; i++) {  
            arr[i] = arr[i + 1];  
        }  
        arr[n - 1] = temp;  
  
        for (int num : arr) {  
            System.out.print(num + " ");  
        }  
    }  
}
```

Answer

2 3 4 5 1

Status : Correct

Marks : 1/1

5. What will be the output of the following code?

```
class Sample {  
    public static void main(String[] args) {  
        int[][] data = {  
            {1, 2},  
            {3, 4}  
        };  
        int sum = 0;  
  
        for (int[] row : data) {  
            for (int val : row) {  
                sum += val;  
            }  
        }  
    }  
}
```

```
        System.out.println("Sum = " + sum);
    }
}
```

Answer

Sum = 10

Status : Correct

Marks : 1/1

6. What will be the output of the following code?

```
class ReverseArray {
    public static void main(String[] args) {
        int[] a = {1, 2, 3, 4};
        for (int i = 0; i < a.length / 2; i++) {
            int temp = a[i];
            a[i] = a[a.length - 1 - i];
            a[a.length - 1 - i] = temp;
        }
        for (int i : a)
            System.out.print(i + " ");
    }
}
```

Answer

4 3 2 1

Status : Correct

Marks : 1/1

7. What will be the output of the following code?

```
class Sample {
    public static void main(String[] args) {
        int[][] matrix = {
            {1, 2, 3},
            {4, 5, 6}
        };
        System.out.println(matrix[1][2]);
    }
}
```

```
}  
}
```

Answer

6

Status : Correct

Marks : 1/1

8. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[] a = {1, 2, 3, 4};  
        for (int i = 0; i < a.length; i++) {  
            if (a[i] % 2 == 0)  
                a[i] = 0;  
        }  
        System.out.println(a[1] + " " + a[3]);  
    }  
}
```

Answer

0 0

Status : Correct

Marks : 1/1

9. What will be the output of the following code?

```
class Sample {  
    public static void main(String[] args) {  
        int[] a = {1, 2, 3};  
        int product = 1;  
        for (int i = 0; i < a.length; i++) {  
            product *= a[i];  
        }  
        System.out.println(product);  
    }  
}
```

Answer

6

Status : Correct

Marks : 1/1

10. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[] nums = {3, 6, 7, 2, 8};  
        int sum = 0;  
        for (int i = 0; i < nums.length; i++) {  
            if (nums[i] % 2 == 0)  
                sum += nums[i];  
        }  
        System.out.println(sum);  
    }  
}
```

Answer

16

Status : Correct

Marks : 1/1

11. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[][] a = {  
            {1, 2},  
            {3, 4}  
        };  
        for (int i = 0; i < a.length; i++) {  
            for (int j = 0; j < a[0].length; j++) {  
                System.out.print(a[i][j] + " ");  
            }  
        }  
    }  
}
```

```
}
```

Answer

1 2 3 4

Status : Correct

Marks : 1/1

12. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[] a = {1, 2, 3, 4};  
        for (int i = 0; i < a.length / 2; i++) {  
            int temp = a[i];  
            a[i] = a[a.length - 1 - i];  
            a[a.length - 1 - i] = temp;  
        }  
        System.out.println(a[0]);  
    }  
}
```

Answer

4

Status : Correct

Marks : 1/1

13. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[] nums = {4, 2, 9, 5};  
        int max = nums[0];  
        for (int i = 1; i < nums.length; i++) {  
            if (nums[i] > max)  
                max = nums[i];  
        }  
        System.out.println(max);  
    }  
}
```


Answer

9

Status : Correct

Marks : 1/1

14. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[][] a = {  
            {1, 2},  
            {3, 4}  
        };  
        int sum = 0;  
        for (int i = 0; i < a.length; i++)  
            for (int j = 0; j < a[0].length; j++)  
                sum += a[i][j];  
        System.out.println(sum);  
    }  
}
```

Answer

10

Status : Correct

Marks : 1/1

15. What will be the output of the following code?

```
class Q {  
    public static void main(String[] args) {  
        int[][] arr = {  
            {5, 6, 7},  
            {8, 9, 10}  
        };  
        System.out.println(arr[0][2]);  
    }  
}
```

Answer

7

Status : Correct

Marks : 1/1

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 3_PAH

Attempt : 1
Total Mark : 40
Marks Obtained : 20

Section 1 : Coding

1. Problem Statement

Eminem is a billiard player who enjoys playing billiards and also likes solving mathematical puzzles. He notices that the billiard balls on the table are arranged in a grid, and he is curious to find the sum of the numbers written on each ball.

Write a program to find the sum of all the numbers written on each ball in the grid.

Input Format

The first line of input consists of an integer N, representing the number of rows.

The second line consists of an integer M, representing the number of columns.

The following lines N lines consist of M space-separated integers, representing the numbers written on each ball.

Output Format

The output prints an integer representing the sum of all the numbers written on each ball.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

3

1 2 3

4 5 6

7 8 9

Output: 45

Answer

-

Status : Skipped

Marks : 0/10

2. Problem Statement

Priya is building a system to automate image transformations using matrix operations. To do this, she needs to multiply two matrices representing pixel data and transformation rules.

Help Priya perform matrix multiplication and print the resulting matrix if the operation is valid.

Input Format

The first line of input consists of two int values, representing the number of rows R1 and columns C1 of the first matrix.

The next $R1 \times C1$ integers represent the elements of the first matrix.

The next line consists of two int values, representing the number of rows R2 and columns C2 of the second matrix.

The next $R2 \times C2$ integers represent the elements of the second matrix.

Output Format

If matrix multiplication is possible, print R1 lines, each containing C2 space-separated int values representing the resulting matrix.

Otherwise, print "Matrix multiplication not possible".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2 3

1 2 3

4 5 6

3 2

7 8

9 10

11 12

Output: 58 64

139 154

Answer

-

Status : Skipped

Marks : 0/10

3. Problem Statement

Egath is participating in a coding hackathon, and one of the challenges requires him to work with an array of integers. The task is to remove exactly one element from the array such that the sum of the remaining elements is a prime number.

Help Egath find the first possible prime sum by removing one element or

determining if no such prime sum can be achieved.

Input Format

The first line of input consists of an integer N, representing the number of elements in the array.

The second line consists of N space-separated integers, representing the array elements.

Output Format

If removing one element results in a prime sum, print the sum.

If no such prime sum can be achieved by removing exactly one element, print "No valid prime sum found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

1 2 3

Output: 5

Answer

```
import java.util.Scanner;

class PrimeSumAfterDeletion {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int n = scanner.nextInt();    // number of elements
        int[] nums = new int[n];
        int totalSum = 0;             // sum of all elements

        for (int i = 0; i < n; i++) { // read array and compute total sum
            nums[i] = scanner.nextInt();
            totalSum += nums[i];
        }
    }
}
```

```

boolean foundPrimeSum = false;    // flag to indicate success

// try deleting each element once
for (int i = 0; i < n && !foundPrimeSum; i++) {
    int remainingSum = totalSum - nums[i];

    // primality test (no extra functions)
    if (remainingSum > 1) {
        boolean isPrime = true;

        if (remainingSum > 3) {
            if (remainingSum % 2 == 0 || remainingSum % 3 == 0) {
                isPrime = false;
            } else {
                for (int j = 5; j * j <= remainingSum; j += 6) {
                    if (remainingSum % j == 0 || remainingSum % (j + 2) == 0) {
                        isPrime = false;
                        break;
                    }
                }
            }
        }
    }

    // remainingSum == 2 or 3 are already prime (isPrime stays true)

    if (isPrime) {          // valid prime sum found
        System.out.println(remainingSum);
        foundPrimeSum = true;
    }
}

if (!foundPrimeSum) {      // no element removal yields a prime sum
    System.out.println("No valid prime sum found");
}
scanner.close();
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

In a customer loyalty program, reward points are logged in a sorted array as customers make transactions. Occasionally, due to system errors, duplicate entries for the same transaction may appear. To ensure accurate reward calculations, it's crucial to remove these duplicates from the list.

Write a program to process the array of reward points, removing any duplicates while preserving the order of unique entries. The program should then display the cleaned list of unique reward points and the total count of these unique points.

Input Format

The first line of input consists of an integer N, representing the number of reward points.

The second line consists of N space-separated integers, representing the reward points in sorted order.

Output Format

The first line of output prints the cleaned list of unique reward points separated by a space.

The second line of output prints an integer representing the total count of unique reward points.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3
100 100 200
Output: 100 200
2

Answer

```
import java.util.Scanner;
```



```

class RemoveDuplicates {

    // Main method to handle input, remove duplicates, and output results
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Input size of array
        int n = scanner.nextInt();

        // Input elements of the array
        int[] nums = new int[n];
        for (int i = 0; i < n; i++) {
            nums[i] = scanner.nextInt();
        }

        // Edge case: If array is empty, handle separately
        if (nums.length == 0) {
            System.out.println();
            System.out.println(0);
            scanner.close();
            return;
        }

        // Index to place the next unique element
        int uniqueIndex = 1;

        // Traverse the array starting from the second element
        for (int i = 1; i < nums.length; i++) {
            if (nums[i] != nums[i - 1]) {
                nums[uniqueIndex] = nums[i];
                uniqueIndex++;
            }
        }

        // Output the cleaned list of unique reward points
        for (int i = 0; i < uniqueIndex; i++) {
            System.out.print(nums[i] + " ");
        }
        System.out.println();

        // Output the total count of unique reward points
        System.out.println(uniqueIndex);
    }
}

```

```
        scanner.close();  
    }  
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Rosh is intrigued by numerical patterns. Today, she stumbled upon a puzzle while working with arrays. She wants to compute the sum of the third-largest and second-smallest elements from a list of integers. She seeks your help to implement a program that solves this for her efficiently.

Input Format

The first line of input is an integer N, representing the size of the array.

The second line of input consists of N space-separated integers, representing the elements of the array.

Output Format

The output displays a single integer representing the sum of the third-largest and second-smallest elements in the array.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 10

10 20 30 40 50 60 70 80 90 100

Output: 100

Answer

```
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int N=sc.nextInt();
        int array1[]=new int[N];
        int array2[]=new int[N];
        int k=0;
        for(int i=0;i<N;i++){
            array1[k]=sc.nextInt();
            k=k+1;
        }
        for (int j = 0; j < array1.length - 1; j++) {
            if (array1[j] < array1[j + 1]) {
                int temp = array1[j];
                array1[j] = array1[j + 1];
                array1[j + 1] = temp;
                j = -1;
            }
        }
        int res=0;
        res=res+array1[2]+array1[array1.length-2];
        System.out.println(res);
    }
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q2

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Monica is interested in finding a treasure but the key to opening is to get the sum of the main diagonal elements and secondary diagonal elements.

Write a program to help Monica find the diagonal sum of a square 2D array.

Note: The main diagonal of the array consists of the elements traversing from the top-left corner to the bottom-right corner. The secondary diagonal includes elements from the top-right corner to the bottom-left corner.

Input Format

The first line of input consists of an integer N, representing the number of rows and columns.

The following N lines consist of N space-separated integers, representing the 2D array elements.

Output Format

The first line of output prints "Sum of the main diagonal: " followed by an integer, representing the sum of the main diagonal.

The second line prints "Sum of the secondary diagonal: " followed by an integer, representing the sum of the secondary diagonal.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

1 2 3

4 5 6

7 8 9

Output: Sum of the main diagonal: 15

Sum of the secondary diagonal: 15

Answer

```
import java.util.Scanner;
```

```
class DiagonalSum {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);
```

```
        // Read size of matrix  
        int n = sc.nextInt();  
        int[][] arr = new int[n][n];
```

```
        // Input the matrix  
        for (int i = 0; i < n; i++) {  
            for (int j = 0; j < n; j++) {  
                arr[i][j] = sc.nextInt();  
            }  
        }
```

```
        int mainDiagonalSum = 0;
```

```
int secondaryDiagonalSum = 0;

// Calculate diagonal sums
for (int i = 0; i < n; i++) {
    mainDiagonalSum += arr[i][i];           // top-left to bottom-right
    secondaryDiagonalSum += arr[i][n - i - 1]; // top-right to bottom-left
}

// Print result
System.out.println("Sum of the main diagonal: " + mainDiagonalSum);
System.out.println("Sum of the secondary diagonal: " +
secondaryDiagonalSum);

    sc.close();
}
}
```

Status : Correct

Marks : 10/10

Rajalakshmi Engineering College

Name: Naren Kartic B
Email: 241901065@rajalakshmi.edu.in
Roll no:
Phone: 6374678252
Branch: REC
Department: CSE (CS) - Section 1
Batch: 2028
Degree: B.E - CSE (CS)

Scan to verify results



2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

You are developing a warehouse management system for a shipping company. The system uses an integer array to represent the weights of packages in a specific order. To verify that the weight capacity is not exceeded, the program needs to calculate the sum of the weights of the first and last packages in the list.

Task:

Write a code to calculate the sum of the weights of the first and last packages in the list. The program should take an integer array as input and return the total weight of the first and last packages.

Input Format

The first line of the input is an integer N representing the size of the array.

The second line of the input is N space-separated integer values.

Output Format

The output is displayed in the following format:

"Sum of the first and last elements: <<Sum>>"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

10 20 30 40 50

Output: Sum of the first and last elements: 60

Answer

```
import java.util.Scanner;
class Jobi{
    public static void main(String[]args){
        Scanner sc = new Scanner (System.in);
        int N = sc.nextInt();
        int []arr = new int [N];
        for(int i =0; i <N ; i++){
            arr[i]= sc.nextInt();
        }
        int sum = arr[0] + arr[N - 1];
        System.out.println("Sum of the first and last elements: " +sum);
        sc.close();
    }
}
```

Status : Correct

Marks : 10/10

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Email: 241901065@rajalakshmi.edu.in
Roll no:
Phone: 6374678252
Branch: REC
Department: CSE (CS) - Section 1
Batch: 2028
Degree: B.E - CSE (CS)

Scan to verify results



2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Sesha is developing a weather monitoring system for a region with multiple weather stations. Each weather station collects temperature data hourly and stores it in a 2D array.

Write a program that can add the temperature data from two different weather stations to create a combined temperature record for the region.

Input Format

The first line of input consists of two space-separated integers N and M, representing the number of rows and columns of the matrices, respectively.

The next N lines consist of M space-separated integers, representing the values of the first matrix.

The following N lines consist of M space-separated integers, representing the values of the second matrix.

Output Format

The output prints the addition of the two matrices in N rows and M columns, representing the combined temperature record.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3 3

1 2 3

4 5 6

7 8 9

1 1 1

2 2 2

3 3 3

Output: 2 3 4

6 7 8

10 11 12

Answer

```
import java.util.Scanner;
class oak{
    public static void main(String[]args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int m = sc.nextInt();

        int [][] matrix1 = new int [n][m];
        int [][] matrix2 = new int [n][m];
        int [][]result = new int[n][m];
        for (int i = 0; i<n;i++){
            for(int j = 0; j<m;j++){
                matrix1[i][j]= sc.nextInt();

            }
        }
        for (int i = 0; i<n ; i++){
```

```
        for(int j = 0 ; j<m ; j++){
            matrix2[i][j] = sc.nextInt();
        }
    }
    for (int i =0 ; i<n;i++){
        for(int j =0;j<m;j++){
            result[i][j] = matrix1[i][j] + matrix2[i][j];

        }
    }
    for (int i =0;i<n;i++){
        for (int j =0 ; j<m;j++){
            System.out.println(result[i][j] + " ");
        }
        System.out.println();
    }
    sc.close();
}
}
```

Status : Correct

Marks : 10/10

Rajalakshmi Engineering College

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Email: 241901065@rajalakshmi.edu.in
Roll no:
Phone: 6374678252
Branch: REC
Department: CSE (CS) - Section 1
Batch: 2028
Degree: B.E - CSE (CS)

Scan to verify results



2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q5

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Sharon is creating a program that finds the first repeated element in an integer array. The program should efficiently identify the first element that appears more than once in the given array. If no such element is found, it should appropriately display a message.

Help Sharon to complete the program.

Input Format

The first line of input consists of an integer n, representing the number of elements in the array.

The second line consists of n space-separated integers, representing the array elements.

Output Format

If a repeated element is found, print the first element that appears more than once.

If no repeated element is found, print "No repeated element found in the array".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 8

12 21 13 14 21 36 47 21

Output: 21

Answer

```
import java.util.Scanner;
import java.util.HashSet;

class FirstRepeatedElement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Read size of array
        int n = sc.nextInt();
        int[] arr = new int[n];

        // Read array elements
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        // HashSet to store seen elements
        HashSet<Integer> seen = new HashSet<>();
        boolean found = false;

        // Traverse array and check for first repeated
        for (int i = 0; i < n; i++) {
            if (seen.contains(arr[i])) {
                System.out.println(arr[i]);
            }
        }
    }
}
```

```
        found = true;
        break;
    } else {
        seen.add(arr[i]);
    }
}

// If no repeated element is found
if (!found) {
    System.out.println("No repeated element found in the array");
}

sc.close();
}
```

Status : Correct

Marks : 10/10